

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a deep blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

LECTURE 9.1

Laminar Flow

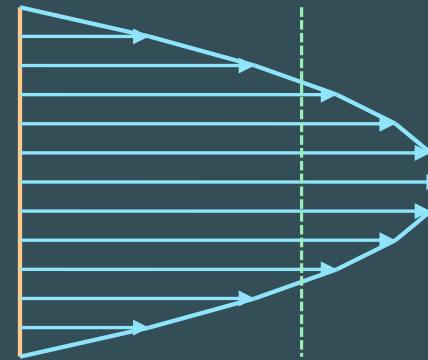
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

NEWTON · THE PIPE · DARCY



fastest on the axis

the mean: half the
centre speed

zero on the wall

Navier–Stokes is Newton's law for a parcel of liquid. In a straight tube far from its bends it reduces to one equation, solved by hand: Hagen–Poiseuille's parabola, Darcy's law, and the start-up when the pump begins. FDM and a PINN then solve it against the truth, and a flow meter gives the viscosity.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Flow equations** — Newton's law for a liquid, the Reynolds number
- 2 **Pipe flow** — fully developed, Hagen–Poiseuille and Darcy, the start-up
- 3 **Solving a flow** — conditions built in, FDM or PINN, the viscosity

White carries the pipe flow and Liu the method; the example is the exercise's own, the coolant in a 6 mm tube of a cold plate.

READ, AND WHAT EACH ANSWERS

White ch. 4	what does Newton say for a liquid?
White ch. 6	what is the flow in a tube?
Liu ch. 2.2	how is it solved, and checked?

Recap of L8.2 Transient Heat

1 – 2 · TIME

1 What does the heat equation add to steady conduction, and how does a PINN treat time?

A storage term, $\rho c_p \partial T / \partial t$, and so one initial field. The diffusivity $\alpha = k / \rho c_p$ says how fast a change spreads and sets the time constant τ . For a PINN time is one more input: the whole block of space and time is trained at once, and the solution can be read at any instant. *Liu ch. 4.1 – 4.3; Ex_08.2 sections 3 – 5*

2 What is causality, and how is it put back into training?

A solution at one instant follows from the earlier ones, but the loss treats all instants alike, so the network can fit late times first and carry an error at the start. Build the initial field into the trial solution and sample early times densely, as Ex_08.2 does; causal weights and a split window go further. *Liu ch. 4.2 – 4.4; Ex_08.2 section 5*

3 – 4 · PLATE AND INVERSE

3 How is the transient plate set up, and how is its solution checked?

The trial solution $T(x, t)$ holds the initial field and the hole wall at once; the insulated edges stay a loss term. With no exact solution, it is scored against a finite-element reference: the worst error over the plate at each instant, and the hottest point in time. *Liu ch. 4.6; Ex_08.2 sections 4 – 5*

4 How does an inverse problem differ from a forward one?

A parameter of the equation, here the diffusivity α , is trained beside the weights so that the solution fits measured data. It is optimised as $\log \alpha$, so it stays positive. Only the early part of the cooling curve constrains it: once the field has decayed, the data says nothing. *Liu ch. 4.6; Ex_08.2 section 6*

Navier-Stokes

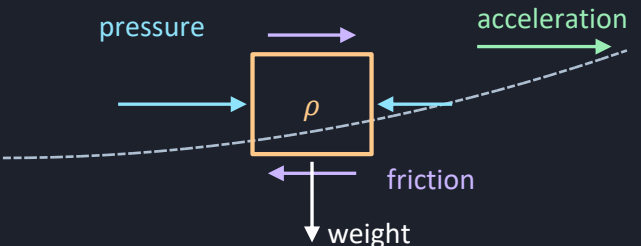
Navier-Stokes — Newton's second law for a small parcel of liquid: mass times acceleration equals the forces on it. The mass per volume is the density ρ ; the acceleration is how the parcel's velocity changes as it moves; the forces are the pressure, the friction of the neighbours, the weight.

MASS TIMES ACCELERATION EQUALS THE FORCES

$$\underbrace{\rho}_{\text{mass}} \underbrace{(\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u})}_{\text{acceleration}} = \underbrace{-\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}}_{\text{the forces}}, \quad \nabla \cdot \mathbf{u} = 0$$

ρ the mass per volume, $\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u}$ the acceleration of the parcel: in time, and because it moves on $-\nabla p$ the push of the pressure, $\mu \nabla^2 \mathbf{u}$ the friction of the neighbours ($\mu = \rho \nu$), $\rho \mathbf{g}$ the weight, $\nabla \cdot \mathbf{u} = 0$ the volume kept

A PARCEL OF LIQUID, AND NEWTON'S LAW ON IT



mass: the density, per volume

acceleration: in time, and as it moves

forces: pressure, friction, weight

NEWTON'S LAW, TERM BY TERM

Newton	for a parcel of liquid	the term
mass	the density, per volume	ρ
acceleration	the velocity changes in time	$\mathbf{u}_t$
	and because it moves on	$(\mathbf{u} \cdot \nabla)\mathbf{u}$
force	the pressure, from the sides	$-\nabla p$
	the neighbours drag on it	$\mu \nabla^2 \mathbf{u}$
	its weight	$\rho \mathbf{g}$

Why $(\mathbf{u} \cdot \nabla)\mathbf{u}$ is an acceleration: in a time dt the parcel moves from x to $x + \mathbf{u} \cdot dt$, where the velocity is different, so its velocity changes by $\mathbf{u}_t \cdot dt + (\mathbf{u} \cdot dt \cdot \nabla)\mathbf{u}$. In a steady flow through a narrowing pipe $\mathbf{u}_t = 0$, yet every parcel speeds up.

Inertia and Darcy

Inertia term — $(\mathbf{u} \cdot \nabla)\mathbf{u}$, the one term where the velocity multiplies itself: the fluid carries its own momentum. Double $\mathbf{u}$ and every other term doubles, this one quadruples. Solutions can no longer be added, large eddies feed smaller ones, and the flow can turn turbulent.

FRAME IT, NEGLECT IT, AVERAGE IT

$$\rho \mathbf{u}_t + \boxed{\rho (\mathbf{u} \cdot \nabla) \mathbf{u}} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

keep it: nonlinear, turbulence

$$\rho \mathbf{u}_t = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

neglect it: Stokes, linear, slow flow

$$\bar{u} = -\frac{K}{\mu} \frac{dp}{dx}, \quad K = \frac{h^2}{12}$$

average across the channel: Darcy
the law this lecture rests on

$(\mathbf{u} \cdot \nabla)\mathbf{u}$ the inertia, the one nonlinear term; $\bar{u}$ the mean speed across the channel, dp/dx the pressure drop per length K the permeability, $h^2/12$ for a channel of height h , μ the viscosity, $\rho \mathbf{g}$ the weight

WHY IT IS NONLINEAR: DOUBLE THE VELOCITY

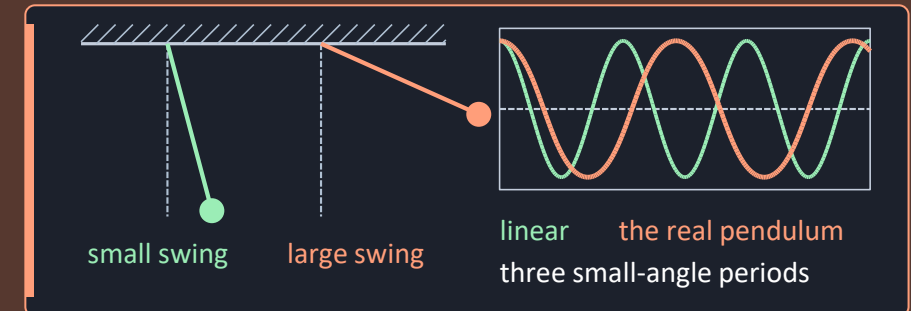
$\rho \mathbf{u}_t, \mu \nabla^2 \mathbf{u}, \nabla p$ double: linear

$\rho (\mathbf{u} \cdot \nabla) \mathbf{u}$ quadruples: $u \cdot u_x = (u^2/2)_x$

THE SAME STEP ON A PENDULUM

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0 \Rightarrow \ddot{\theta} + \frac{g}{\ell} \theta = 0$$

θ the angle, ℓ the length, g gravity; $\sin \theta \rightarrow \theta$ is the same step as dropping the inertia



Small swing, $\sin \theta \approx \theta$: linear, the same period at any amplitude - that is Stokes. Large swing: the period grows with the amplitude, driven it goes chaotic - that is Navier–Stokes. The amplitude decides whether the nonlinearity matters, as Re does for a flow.

The Reynolds Number

Reynolds number — the ratio of inertia to viscosity, $Re = \rho \bar{u} D / \mu$. In a pipe below about 2300 the flow stays laminar, layer sliding over layer; above it, it turns turbulent. It also sets how far the flow needs after a bend to settle into its profile: the entrance length, $0.05 Re D$.

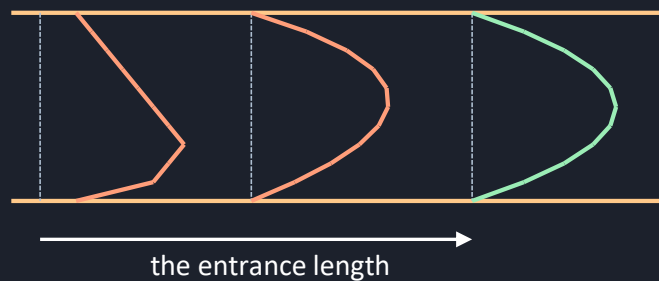
ONE NUMBER, AND THE LENGTH IT SETS

$$Re = \frac{\rho \bar{u} D}{\mu} = 452$$

$$L_e \approx 0.05 Re D = 136 \text{ mm}$$

ρ the density, $\bar{u}$ the mean speed, D the diameter, μ the viscosity
 L_e the entrance length: how far after a bend the profile needs to settle

AFTER A BEND, THE PROFILE SETTLES

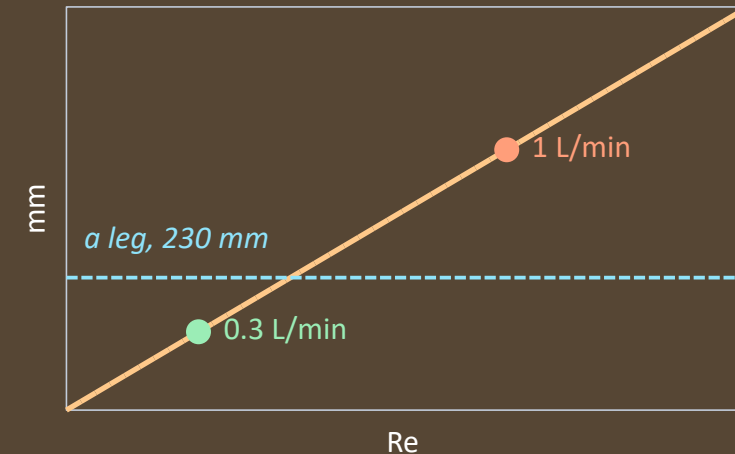


after a bend: skewed

settling along the leg

the parabola, fully developed

ENTRANCE LENGTH AGAINST Re



The cold plate's 6 mm tube. At 0.3 L/min the parabola forms 136 mm after each bend, inside a 230 mm leg; at 1 L/min it would need 452 mm and never form.

Flow Boundary Conditions

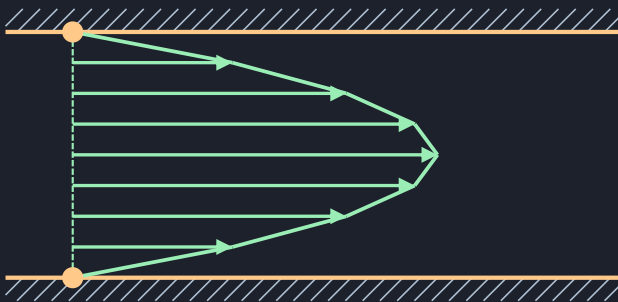
Flow boundary conditions — a wall carries no-slip, $\mathbf{u} = 0$: the fluid touching it moves with it, stuck not slowed - an assumption, confirmed by experiment. In a tube $u = 0$ at $r = R$; on the axis the profile is symmetric, so $u_r = 0$; at the start the coolant is at rest; the pump sets G .

WALL, AXIS, START, PUMP

$$u(R, t) = 0, \quad u_r(0, t) = 0, \quad u(r, 0) = 0$$

u the velocity along the tube, r the distance from the axis, R the radius
 u_r its slope across: zero on the axis, where the profile is symmetric

ACROSS THE TUBE, WALL TO WALL

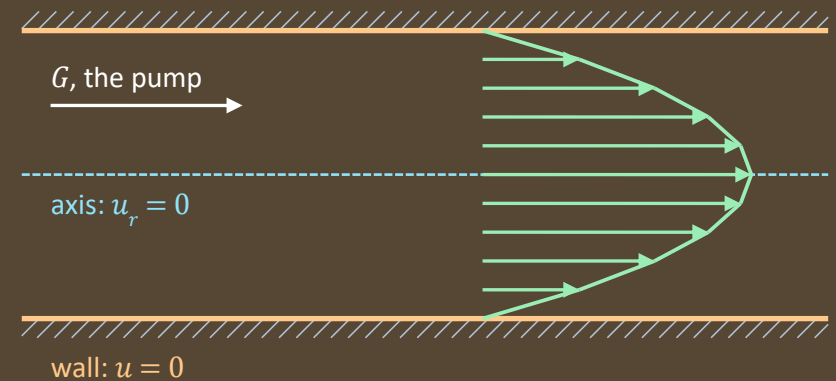


on a wall the fluid does not move

so the speed rises from zero

the same on both walls

THE TUBE OF EX_09.1, ALONG ITS AXIS



wall	$u = 0$: no slip, stuck not slowed
axis	$u_r = 0$: symmetry
start	$u = 0$: at rest
pump	G held from $t = 0$

Fully Developed Flow

Fully developed flow — the flow in a straight tube far from its bends, where the profile no longer changes along the tube. Keeping the volume then forces the velocity towards the wall to zero, the inertia term is exactly zero, and only $u(r, t)$ along the tube is left: Stokes is the full equation.

STOKES ALONG THE TUBE, IN THREE LINES

$$\rho \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \nabla^2 u, \quad \frac{\partial p}{\partial x} = -G$$

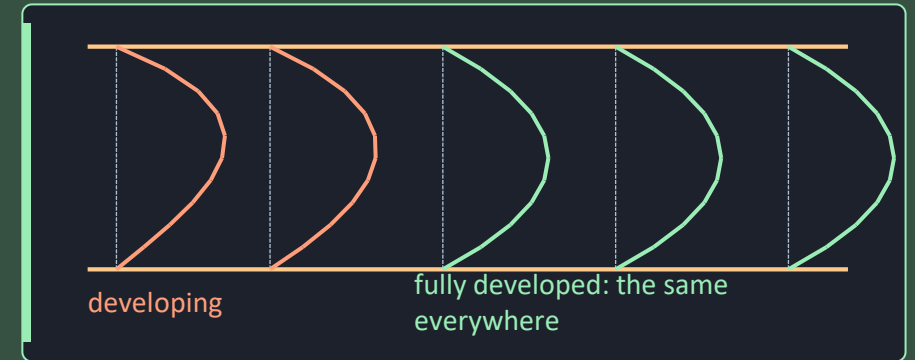
$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$\rho \frac{\partial u}{\partial t} = G + \mu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

u the velocity along the tube, x along, r from the axis, θ round it, ∂ a partial derivative
 G the pressure gradient, the drop per metre (Pa/m), μ the viscosity, ρ the density

Line 1, the x part of Stokes: p falls evenly along a straight tube, by G per metre, so $-\partial p / \partial x = G$. Line 2, the Laplacian in tube coordinates: the first term is zero (fully developed), the last is zero (round: the same at every angle θ). Line 3, what is left.

DEVELOPING, THEN DEVELOPED



round, straight

the same at every θ : no swirl

far from bends

$\partial u / \partial x = 0$: fully developed

volume kept

then $v = 0$: only $u(r, t)$

Hagen–Poiseuille Flow

Hagen–Poiseuille flow — the steady flow in a fully developed tube: a parabola, fastest on the axis and zero on the wall. Its mean speed is Darcy's law with the permeability of a round tube, $K = R^2/8$, exact here because the inertia is zero; written as a friction factor, $f = 64/Re$.

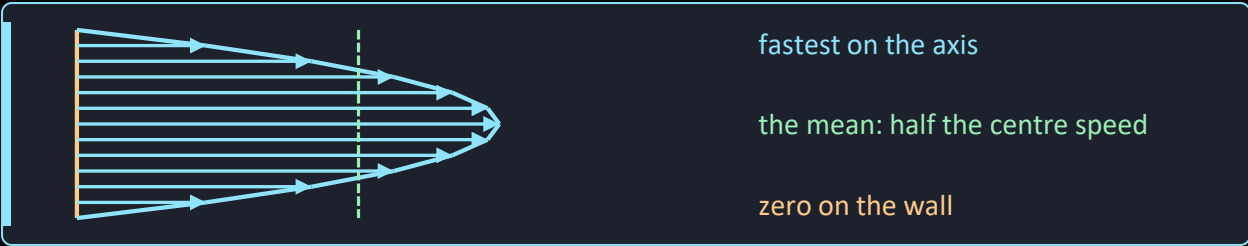
THE PARABOLA, ITS MEAN, ITS FRICTION

$$u(r) = \frac{GR^2}{4\mu} \left(1 - \frac{r^2}{R^2}\right)$$

$$f = \frac{64}{Re}$$

$$\bar{u} = \frac{GR^2}{8\mu} = \frac{K}{\mu} G, \quad K = \frac{R^2}{8}$$

R the tube's radius, G the pressure gradient (Pa/m), $\bar{u}$ the mean speed, K the permeability of a round tube
 f Darcy's friction factor, Re the Reynolds number



THE COLD PLATE'S TUBE, STEADY

quantity	value
mean speed $\bar{u}$	0.177 m/s
centre speed	0.354 m/s, twice the mean
pressure gradient G	393 Pa per metre
friction $f = 64/Re$	0.142
permeability K	1.1 mm ²

Why a parabola: the pressure pushes every shell alike. The friction between shells carries the push of everything inside out to the wall, so the shear stress grows with r , $\tau = Gr/2$, and u falls from the axis to zero at the sticking wall. Without friction every parcel would go alike.

Start-Up Flow

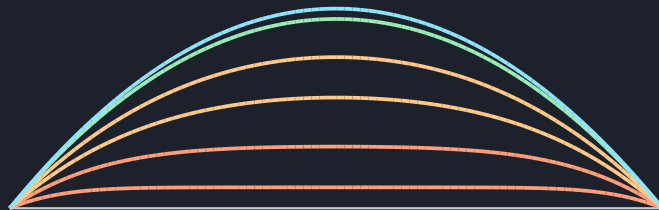
Start-up flow — the same tube from rest, when the pump starts. Its exact solution is the parabola less a sum of terms that each die away at their own rate. Hagen–Poiseuille is not a second answer: it is where the start-up ends, after about $R^2/\nu = 3.8$ s.

THE PARABOLA, LESS WHAT HAS NOT ARRIVED

$$u(r, t) = u(r) - \frac{GR^2}{4\mu} \sum_n c_n J_0 \left(\frac{\lambda_n r}{R} \right) e^{-\lambda_n^2 \nu t / R^2}$$

$$c_n = \frac{8}{\lambda_n^3 J_1(\lambda_n)}, \quad J_0(\lambda_n) = 0$$

J_0, J_1 Bessel functions, λ_n the zeros of J_0 : 2.405, 5.520, 8.654
 $\nu = \mu/\rho$ the kinematic viscosity, $u(r)$ the parabola, c_n the weight of each term

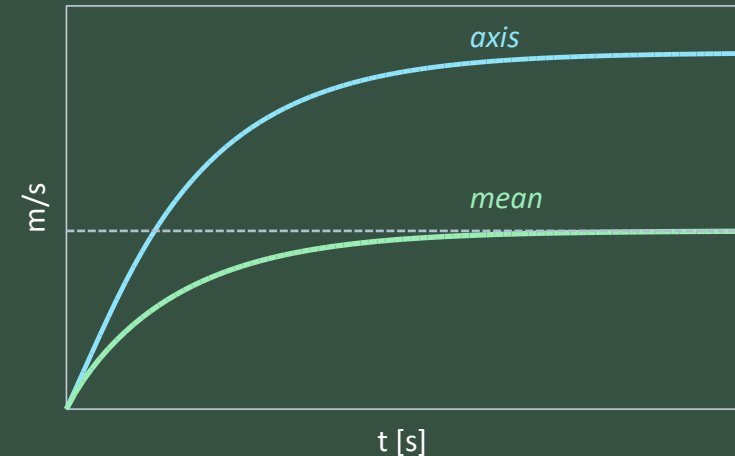


0.1 s: a flat core, pushed at once

1 s: rounding

4 s: the parabola

THE MEAN SPEED, AND THE AXIS



The mean speed - what a flow meter reads - is half-way after 0.43 s and within a per cent after about three.
 Ex_09.1's exact solution.

Built-In Conditions

Built-in conditions — the wall and the start are put into the trial function as factors, $(1 - s) \hat{t}$, and the symmetry about the axis by taking $s = r^2/R^2$ as the input. Whatever the PINN does, all three hold, so the loss is the equation alone: hard enforcement.

THE SCALED EQUATION, AND THE TRIAL FUNCTION

$$\hat{u}_{\hat{t}} = a + 4\kappa (s \hat{u}_{ss} + \hat{u}_s)$$

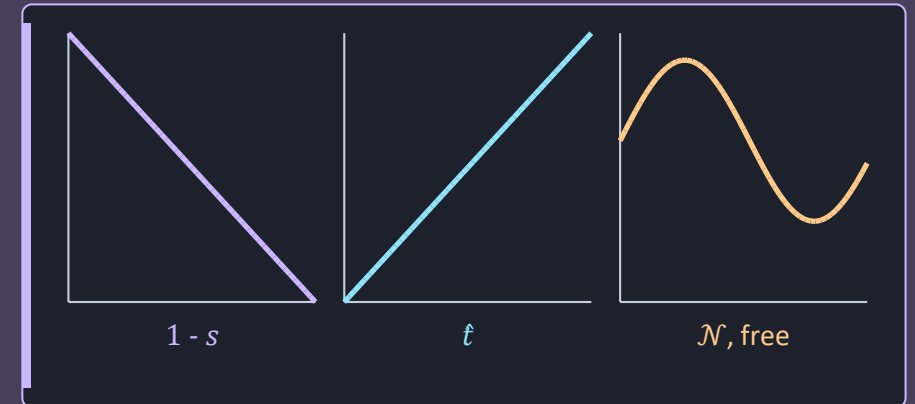
$$\hat{u} = (1 - s) \hat{t} \mathcal{N}(s, \hat{t})$$

$s = r^2/R^2$ the radius, $\hat{t} = t/4s$, $\mathcal{N}$ the network, κ the viscosity and a the push, scaled

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)      # s and t in
u = (1 - s) * t * net(torch.cat([s, t], 1))      # wall and start, built in
us = grad(u, s); ut = grad(u, t)                # derivatives by autograd
r = (ut - a) / kappa - 4 * (s * grad(us, s) + us) # the residual, the whole loss
```

ONE FACTOR FOR EACH CONDITION



$1 - s$	zero on the wall
$\hat{t}$	zero at the start
input s	symmetric about the axis

FDM or PINN

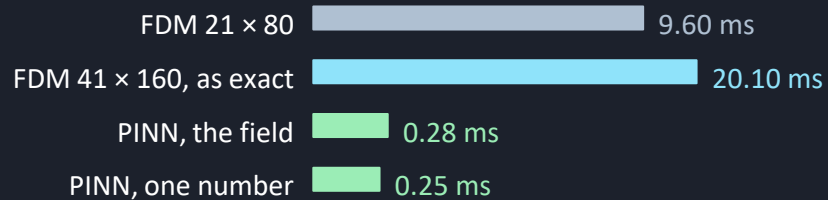
FDM or PINN — finite differences (FDM) compute every answer from scratch. A PINN pays once, offline, for its training, and then answers online in a fraction of a millisecond. It is worth it after N^* answers, or where FDM grows out of reach: many queries, and many dimensions.

WHEN THE TRAINING IS REPAID

$$N^* = \frac{T_{\text{train}}}{t_{\text{FDM}} - t_{\text{net}}}$$

T_{train} the training, t_{FDM} one FDM run, t_{net} one answer of the trained network
 N^* the number of answers after which the training is repaid

ONE GPU, THE SAME ACCURACY, LOG SCALE



trained offline in about 100 s: repaid after about 5 000 answers

HOW THE COSTS GROW

box	FDM, one answer	PINN
1-D, 41	0.35 s	13 ms an iteration
2-D, 41	0.95 s	16 ms an iteration
3-D, 41	1.90 s	26 ms an iteration
3-D, 161	about 1 845 s	about 92 s training

Measured on a laptop GPU in Ex_09.1; the last row is extrapolated, and it holds only if the PINN reaches the same accuracy - which has to be checked.

Inverse Viscosity

Inverse viscosity — the coolant's viscosity found from a flow meter's record. Any model that predicts the readings can be fitted to them: the exact series, FDM in an optimiser, or a PINN with the viscosity trained beside its weights.

ONE MORE TRAINABLE NUMBER

$$\kappa = e^\beta, \quad \mathcal{L} = \mathcal{L}_{PDE} + w_d \sum_k \left(\int_0^1 \hat{u} ds - \bar{u}_k \right)^2$$

β the trained number, κ the viscosity, $\bar{u}_k$ the flow meter's readings, w_d their weight

THE INVERSE IN PYTORCH

```
beta = torch.nn.Parameter(torch.tensor(0.0))      # kappa = exp(beta), not an output
opt = torch.optim.Adam([*net.parameters(), beta]) # trained with the weights
loss = residual(net, beta.exp()).pow(2).mean() + w_d * misfit # PDE + readings
```

FOUR WAYS, ONE RECORD (TRUE 3.50)

method	mPa·s	cost
Darcy, last reading	4.04	none
the exact series	3.55	instant
FDM, optimised	3.53	5 runs, 12 ms
the PINN	3.56	46 s

Ex_09.1, the first second of the record. Darcy is 15 % high because the flow has not settled. An inverse problem is not a PINN's privilege; it pays where no cheap solver exists.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
fronts, Reynolds ceiling	a thin feature against the points	Liu ch. 5.6	mixing layers, high Re
stream function	continuity built in exactly	Liu ch. 2.5.3	2-D flows past bodies
pressure reference	the level the equations leave free	Liu ch. 2.2	any closed duct
bends: Dean flow	the vortices of a curved tube	White ch. 6	serpentines, MP9.1A
<u>porous media</u>	Darcy flow through a solid	Liu ch. 2.4	groundwater, CO ₂ storage
turbulent pipe flow	an averaged, modelled flow	Pope ch. 7	Re above 2300

WHY THEY ARE LEFT OUT

Ex_09.1's tube is straight, laminar and fully developed, with an exact solution to score against, so none of them is needed to do it. The fourth row is mini project MP9.1A, where the students build the solver for a bend themselves.

Exercise 9.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the cold plate, the physics, the exact start-up
- **4** — the exact solution, and finite differences
- **5** — a PINN, conditions built in; timed at equal accuracy
- **6** — the coolant's viscosity from a flow meter
- **7 – 9** — the report; mini projects MP9.1A and MP9.1B

THE QUESTION EACH PART ANSWERS

- 1 – 3** what is left of Navier–Stokes?
- 4** which grid is fine enough?
- 5** when does a PINN pay?
- 6** which way finds the viscosity?

REPORT QUESTION, TO CONCLUDE

At the same accuracy the trained PINN answered 70 times faster than FDM on one GPU, but only 5 times faster than SciPy on a CPU. Where in this cold plate would you choose the PINN, and what would you check first?

Key Takeaways

PIPE FLOW

what it is	Newton's law in a straight tube
function	the velocity across it, in time
train, tune	exact: parabola, Darcy, start-up
vs a bend	no inertia, no vortices
used for	pressure drop, pump, flow rate

SOLVING A FLOW

what it is	a PINN, conditions built in
function	$u(s, t)$, and κ from data
train, tune	the equation alone; then L-BFGS
vs FDM	slower once, faster after N^*
used for	many queries, many dimensions

Next — L9.2: a gas in a fuel-cell channel, and what it carries.

Questions

1 – 2 · PIPE FLOW

- 1 **What is left of Navier–Stokes in a straight tube far from its bends, and why?**
Only the velocity along the tube, changing across it and in time. Round and straight, nothing makes it swirl; far from the bends, nothing changes along the tube; keeping the volume then makes the velocity towards the wall zero, so the inertia term vanishes. *White ch. 6; Ex_09.1 section 3*

- 2 **What do Hagen–Poiseuille and Darcy's law say, and when do they hold?**
The steady profile is a parabola, and the mean speed is the pressure drop times $R^2/8\mu$: Darcy's law with $K = R^2/8$, or $f = 64/\text{Re}$. They hold for laminar flow, Re below about 2300, beyond the entrance length $0.05 \text{ Re } D$ after a bend. *White ch. 6; Ex_09.1 section 3*

3 – 4 · SOLVING A FLOW

- 3 **Which conditions can be built into a PINN for the tube, and how?**
The wall and the start by factors in the trial function, $(1 - s) t$, and the symmetry about the axis by taking $s = r^2/R^2$ as the input. Whatever the weights, all three hold, so the loss is the equation alone. *Liu ch. 2.2; Ex_09.1 section 5*

- 4 **When is a PINN worth more than FDM, and how is a viscosity found from data?**
When many answers are wanted or FDM grows out of reach: a trained PINN answers fast but repays its training only after many runs. The viscosity is fitted to a flow meter's readings by the exact series, FDM in an optimiser or a network. *Ex_09.1 sections 5 – 6*

NEXT

L9.2 — Gas Flow

A gas in the fuel channel of a solid oxide cell: the flow is known, and the question is what it carries - hydrogen in, steam out - along the channel.

BEFORE THE NEXT SESSION

read	White ch. 6
run	Ex_09.1, sections 1 to 7
bring	your viscosity from the flow meter
think about	where you need many answers

*A laminar flow, solved exactly and by two methods.
Next, a flow that carries something along with it.*